

11/11/22

Roll No.

TBT-303

B. TECH. (BIOTECHNOLOGY)

(THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

MICROBIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Give the structural organization of prokaryotic cell wall using suitable diagram. (CO1/CO2)

OR

- (b) Explain how the murein layers are arranged in bacterial cell wall that makes the cell rigid. (CO1/CO2)
2. (a) Using suitable diagram, give the shape and arrangements found in a spherical bacterial cell. (CO1)

OR

- (b) Give any method used to demonstrate the shape of bacterial cell in Microbiology lab. Make suitable diagram. (CO1)

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(2)

3. (a) How does the Gram positive bacteria differ from Gram negative bacteria ? (CO1)

OR

- (b) List the different organelles present the prokaryotic cell. Explain the function of any *one*. (CO1)
4. (a) Why the acid fast bacteria are so called ? How can it be identified in the Microbiology lab using staining method ? (CO1)

OR

- (b) A test-tube containing both yeast culture and bacterial cocci is provided to you. How can it be differentiated under microscope ? Give a brief process used for their identification. (CO1)
5. (a) How the biogenesis theory and abiogenesis theory were different ? Which theory was disapproved finally ? (CO2)

OR

- (b) What do you mean by Golden Age of Microbiology ? Explain briefly. (CO2)